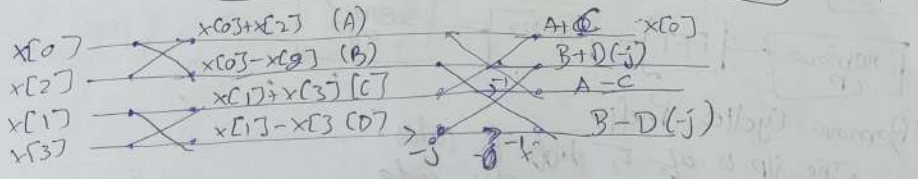
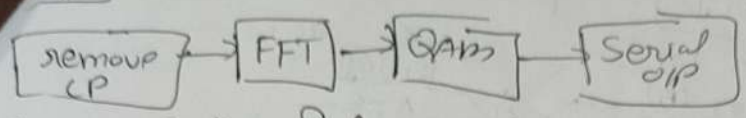
X[0]	X[1]	X[2]	X3
A + C	B + D(-j)	A - C	B - D(-j)



realme Shot on realme GT NEO 3T
2026 04 11 18:14



① Remove Cyclic Prefix

- The i/p is of 5 diff codes.
- We are removing the 1st code

② FFT:

DFT : $X[K] = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi Kn/N}$

FFT : for $N=4 \rightarrow$ ~~16~~ 8 operation.

① $N=4$, split into Even & Odd samples $x[0], x[1], x[2], x[3]$

Even : $x[0], x[2]$

Odd : $x[1], x[3]$

Concept :

we now have 4 sound sample $(x(0), x[1], x[2], x[3])$

FFT asks which combination of sin wave would produce these samples exactly.

$\Rightarrow$ $\boxed{x[0]}$ $\boxed{x[1]}$ $\boxed{x[2]}$ $\boxed{x[3]}$

split into odd & even

$\boxed{x[0]}$ $\boxed{x[2]}$ $\boxed{x[1]}$ $\boxed{x[3]}$

butterfly stag

$\boxed{A \quad x[0]+x[2]}$ $\boxed{B \quad x[0]-x[2]}$ $\boxed{A\ C \quad x[1]+x[3]}$ $\boxed{B\ D \quad x[1]-x[3]}$

apply twiddle fact to D only.
(multiply $-j$)

$\boxed{D(-j) \quad \text{real} = +Dr \quad \text{imag} = -Dr}$

realme Shot on realme GT NEO 3T 2026 04 11 18:14

→ what's the actual output of the signal in QAM?

- we have a output of $I + jQ$ ⇒ which of 16 bit in each,

$Q \rightarrow$ Phase $\rightarrow \tan^{-1} \frac{B_K}{A_K}$ $\quad I \rightarrow$ amplitude $\rightarrow \sqrt{A_K^2 + B_K^2}$

→ How OFDM comes into play?

The $I + jQ$ is of single frequency, where we will have 4 different sub-carrier, ~~we~~

- The 1st & 2nd subcarrier will carry the QAM o/p.
- The 3rd & 4th " " have 0.

↳ 4 point IFFT.

→ Next comes IFFT:

↳ $x[n] = \frac{1}{N} \sum_{K=0}^{N-1} x(K) W_n^{-nK}$ $\quad \left(W_n = e^{\frac{j2\pi}{N}}\right)$

twiddle factor for $N = 4$

$W_4^0 = e^{j\frac{2\pi}{4}(0)} = 1$

$W_4^1 = e^{j\frac{2\pi}{4}(1)} = j$

$W_4^2 = e^{j\frac{2\pi}{4}(2)} = -1$

$W_4^3 = e^{-j\frac{2\pi}{4}(3)} = -j$

00	00	0
01	10	2
10	01	1
11	11	3

DIT

00 →	00	(0)
01 →	10	(2)
10 →	01	(1)
11 →	11	(3)

Stage 1:

Butterfly A = i/p $x[0]$ & i/p $x[2]$

Butter B = i/p $x[1]$ & i/p $x[3]$

top = top_in + bottom_in

bottom = top_in − bottom_in

realme Shot on realme GT NEO 3T 2026 04 11 18:13

$x_0 = 0$
$x_1 = 0.707$
$x_2 = 1$
$x_3 = 0.707$
$x_4 = 0$
$x_5 = -0.701$
$x_6 = -1$
$x_7 = -0.707$

$$X_K = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi n K}{N}}$$

Kth frequency

↳ $X_0, X_1 \cdots X_{N-1}$ => are individual frequency bins that makes up the original wave

→ Fast fourier ~~Sample~~ Transform

$$F_K = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi K n}{N}}$$

↳ DFT which has complexity of N^2

$$\boxed{F_K = \sum_{n=0}^{N-1} x_n e^{-\frac{j2\pi K n}{N}}}$$

By splitting the periodicity into odd & even terms.

$$F_K = \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{j2\pi K(2m)}{N}} + \sum_{m=0}^{\frac{N}{2}-1} x_{2m+1} e^{-\frac{j2\pi K(2m+1)}{N}}$$

↳ Given is $\boxed{O(n\log(n))}$

$$W_N = e^{-j\frac{2\pi}{N}}$$

1) Complex conjugate symmetry $= W_N^{K(N-n)} = W_N^{-Kn} = (W_N^{Kn})^*$

2) Periodicity $W_N^{Kn} = W_N^{K(N+n)} = W_N^{(K+N)n}$

realme Shot on realme GT NEO 3T 2026 04 11 18:13

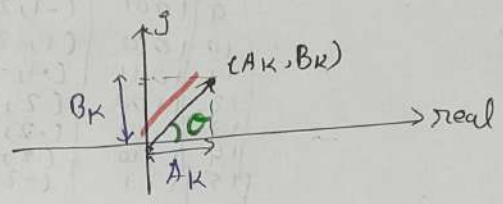
· Discrete Fourier Transform

$$X_K = \sum_{n=0}^{N-1} x_n \cdot e^{\frac{-j2\pi Kn}{N}} \qquad \left(F = \frac{K}{N},\ n = t\right)$$

⇓

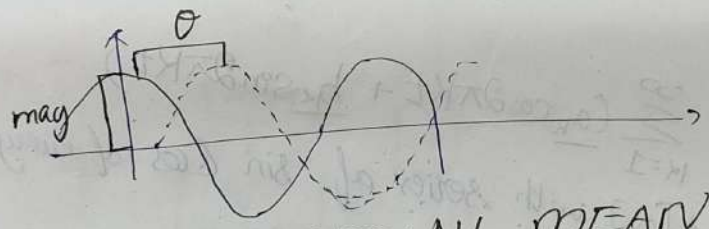
we end up with, which can be plotted in Complex plane

$$X_K = A_K + jB_K$$



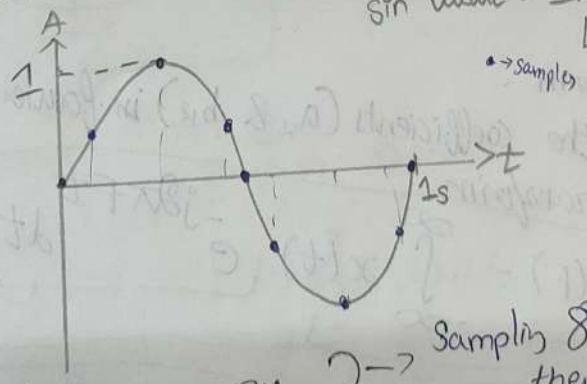
— = magnitude = $\sqrt{A_K^2 + B_K^2}$

θ = Phase = $\tan^{-1}\frac{B_K}{A_K}$



⇒ WHAT DOES THIS ALL MEAN ?

sin wave = 1Hz,
↳ Amplitude = 1

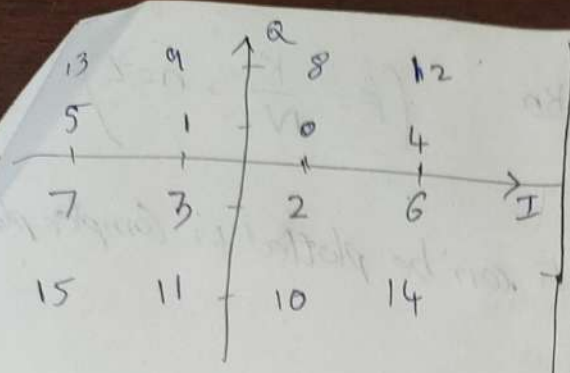


Sampling frequency = 8Hz
Samples (N) = 8

→ Sampling 8Hz for 1s then N = 8

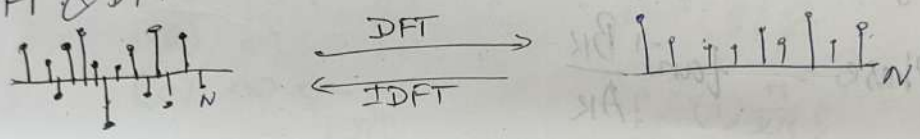
Sampling 8Hz for 10s = then N = 8 × 10 = 80

realme Shot on realme GT NEO 3T 2026 04 11 18:13



0	0000	(1,1)
1	0001	(-1,1)
2	0010	(1,-1)
3	0011	(-1,-1)
4	0100	(2,1)
5	0101	(-2,1)
6	0110	(2,-1)
7	0111	(-2,-1)
8	1000	(1,2)
9	1001	(-1,2)
10	1010	(1,-2)
11	1011	(-1,-2)
12	1100	(2,2)
13	1101	(-2,2)
14	1110	(2,-2)
15	1111	(-2,-2)

→ FFT & DFF:



· Fourier Series:

$$f(t) = \frac{1}{2}a_0 + \sum_{k=1}^{\infty} (a_k \cos 2\pi k t + b_k \sin 2\pi k t)$$

Sum of a constant $[a_0]$ with series of sin & cos of every every frequency.

· Fourier Transform:

To find the coefficients (a_k & b_k) in fourier Series, we use fourier Transform.

Result: 1 complex co-efficient per frequency. ← $X(F) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi F t}\, dt$ analyzing fn.

$$X_a(F) = \int_{-\infty}^{\infty} x(t) \cos 2\pi F t \, dt$$

$$X_b(t) = \int_{-\infty}^{\infty} x(t) \sin 2\pi F t \, dt$$

Resut: 2 real co-efficients per frequency

realme Shot on realme GT NEO 3T 2026 04 11 18:13

→ 16 bit QAM:
16 possible points ($2^4 = 16$)

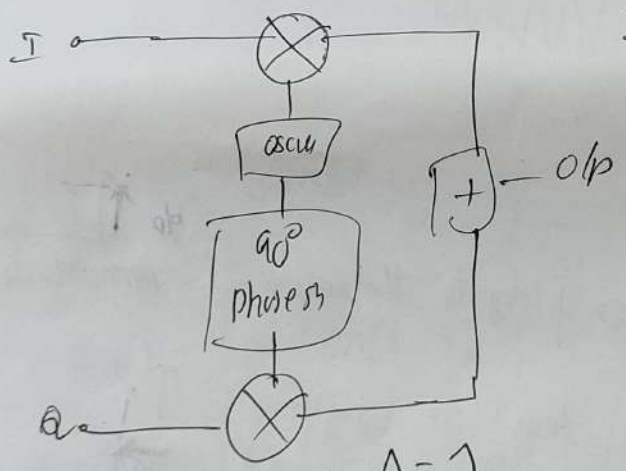
BPSK → 0° & 180° for representing 0 & 1

QPSK → 0°, 90°, 180°, 270° → (00, 01, 10, 11) → 2 bits per symb

└ 4 QAM

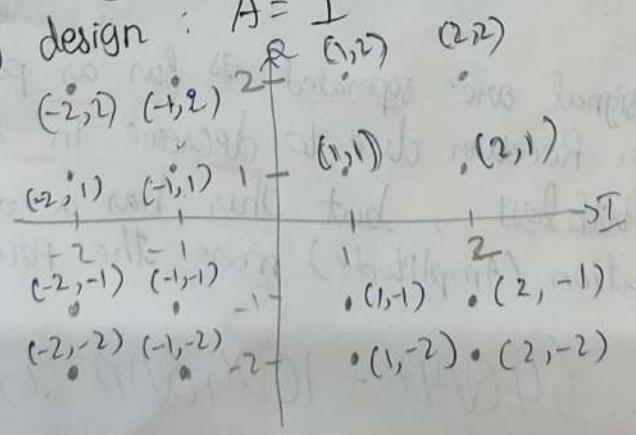
16 QAM → 16 different waveforms

mod	Bits / Symbols	Symbol rate
BPSK	1	1
QPSK	2	1/2 bit/rate
8PSK	3	1/3
16 QAM	4	1/4 → 2 → 10.5



I ← in-phase
Q ← quadrature.

→ for my design : A = 1



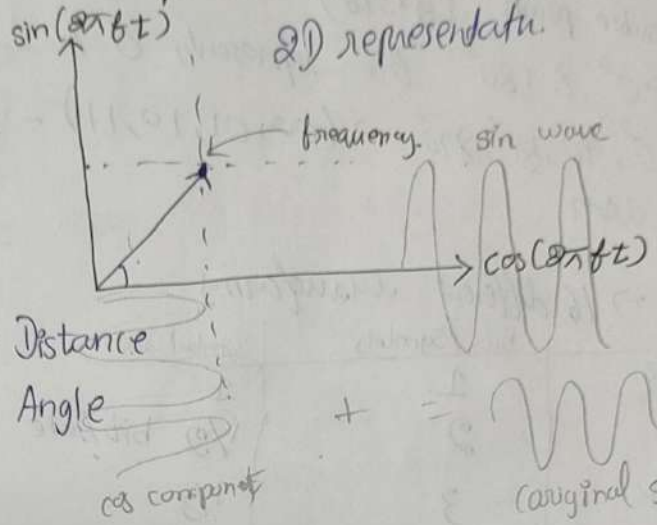
realme Shot on realme GT NEO 3T 2026 04 11 18:12

(1) (2)

(Quadrature Amplitude modulation):

modulating both amp & phase

2D representation

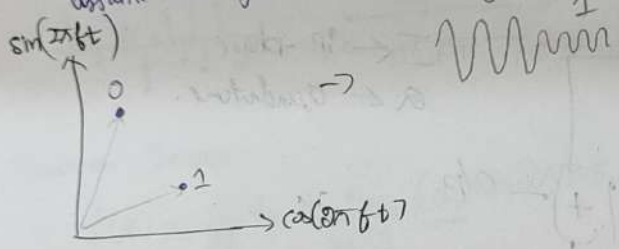


Amplitude ← Distance

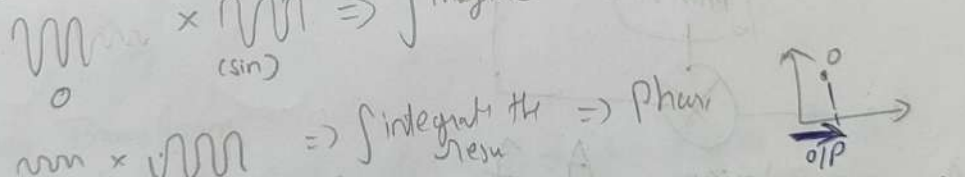
Phase ← Angle

→ TX:

assume we got 2 signals 0 and 1



→ RX:



*The message signal are seperated as far as possible to avoid errors in Receiver due to decrease in Amplide caused by Pathloss, but this has a drawback more the seperation (Amplitude) more the power requires to transmitt.

64QAM - 1024QAM is used in 5G

*How to implement OFDM TX:

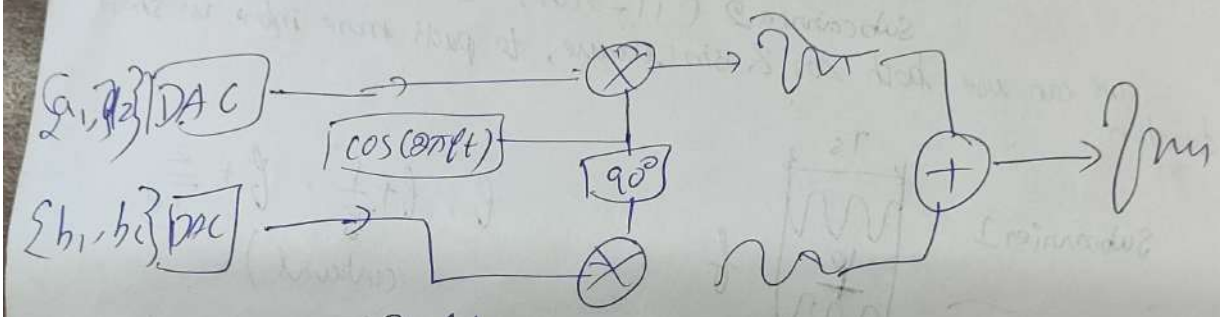
for every data [10, 10, 11, 01] we TX will have different amplitude, & represent them as complex numbers.

$1+j1$ $\quad$ $1+j1$

$\underbrace{1}_{\cos}+\underbrace{j1}_{\sin}$

$$1+j1 \xrightarrow{IFFT} \underbrace{a_1}_{real} + j\underbrace{b_1}_{imagn} \longrightarrow \{a_1\} \xrightarrow{DAC} \sim$$

$$\{b_1\} \xrightarrow{DAC} \sim$$



$$= \text{Real}\left(e^{j2\pi ft} W_x\right)$$

$$= \text{Real}\left[\left[\cos(2\pi ft) + j\sin(2\pi ft)\right] \cdot \text{Real}(W_x + j\,\text{Img}(W_x))\right]$$

$$= \cos(2\pi ft) \cdot \text{Real}(W_x) - \sin(2\pi ft) \cdot \text{Img}(W_x)$$

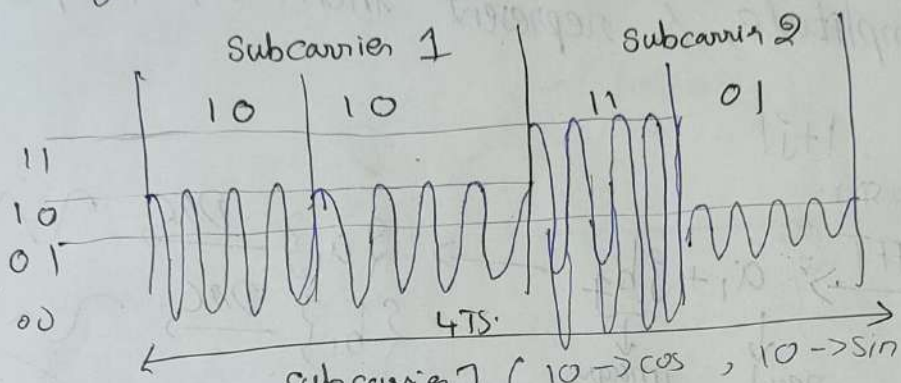
5G uses 4096 subcarriers.

realme Shot on realme GT NEO 3T 2026 04 11 18:12

5G - OFDM

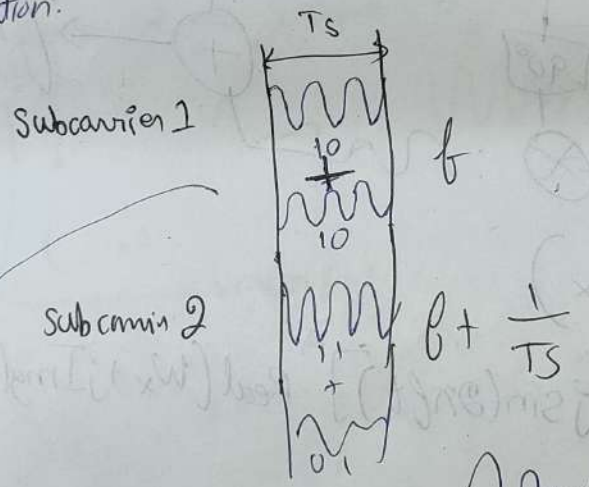
(1)

→ Message signal → 10, 01, 11, 10



↳ OFDM → Subcarrier 1 (10 → cos , 10 → sin)
+
Subcarrier 2 (11 → cos , 01 → sin)

we can use both cos & sin wave, to pack more info in short duration.



$f,\ f+\frac{1}{T_s},\ f+\frac{2}{T_s}$ (interval)

Final o/p of Tx :

* Every wave which we use in subcarrier 1 & 2 are orthogonal.

10 × → other (10, 11, 01 becomes 0)
10 → 10 " "
11 → 11 "
01 → 01